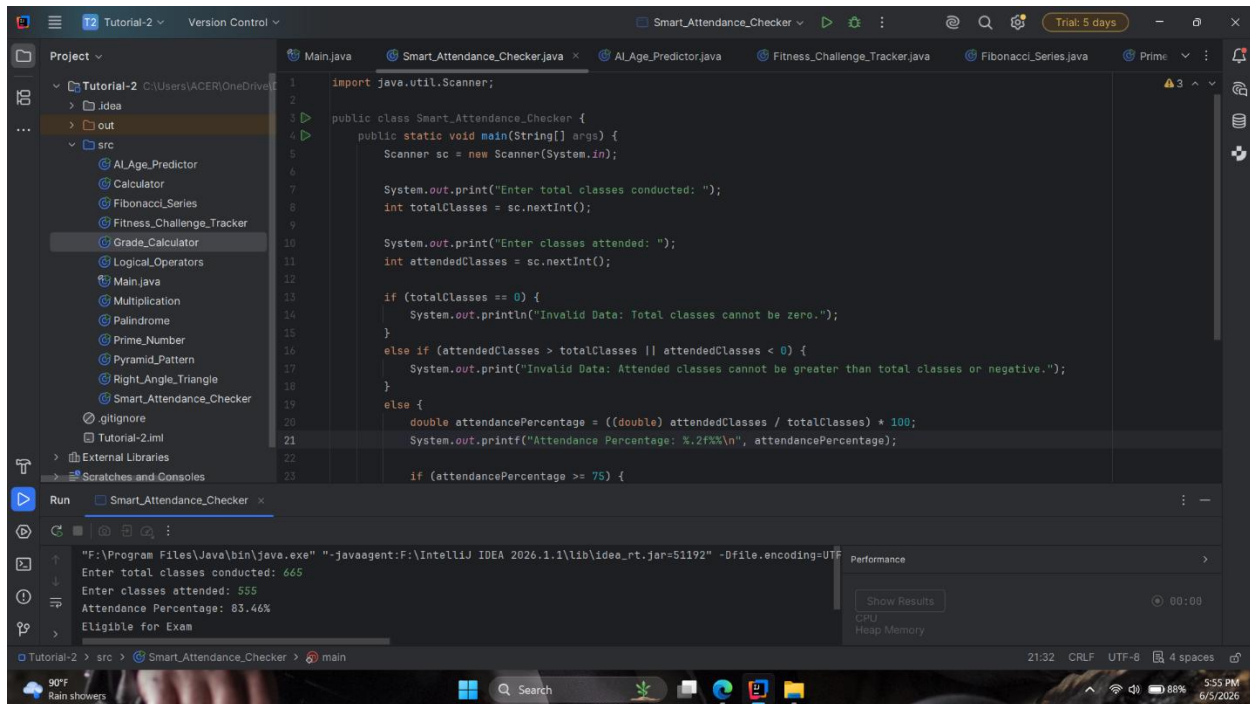


Smart Attendance Checker



The screenshot shows the IntelliJ IDEA IDE with the 'Smart Attendance Checker' project open. The 'Main.java' file is displayed, containing the following code:

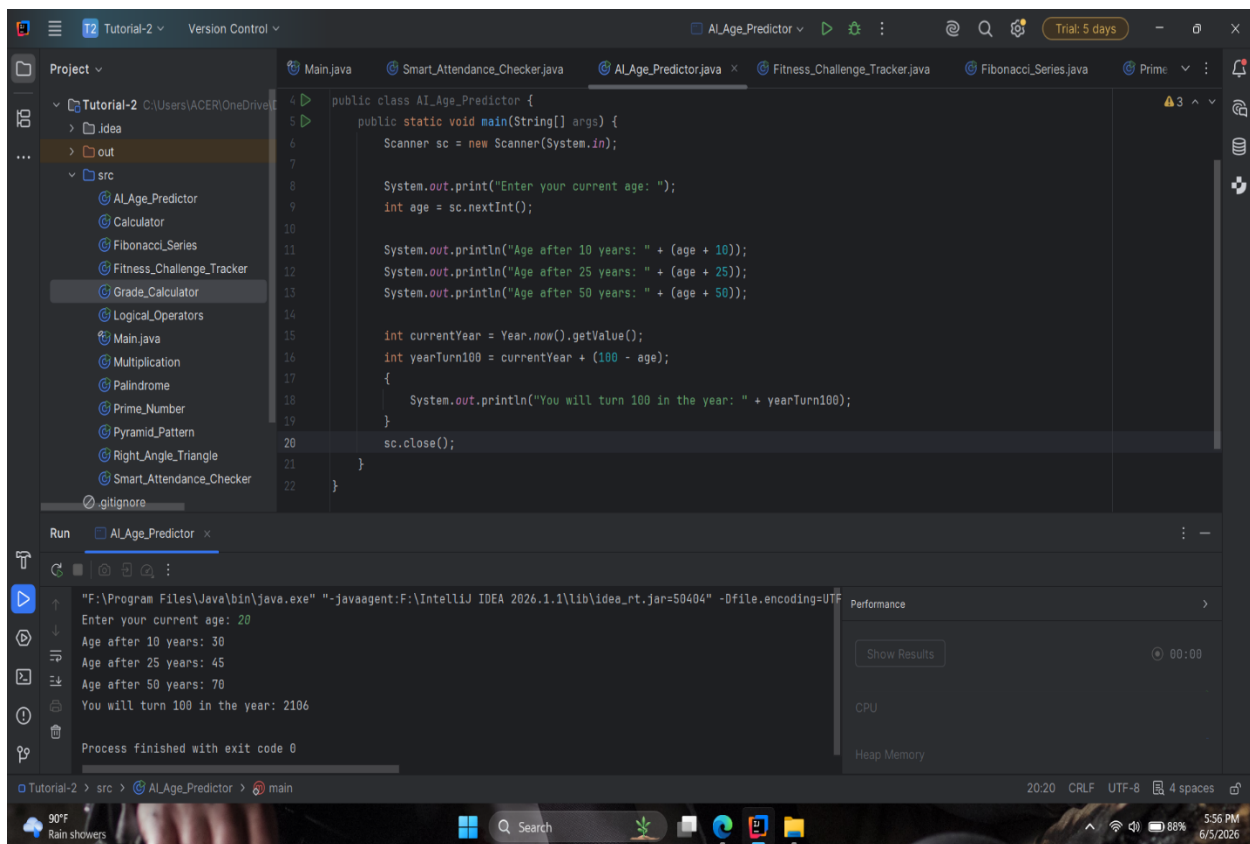
```
1 import java.util.Scanner;
2
3 public class Smart_Attendance_Checker {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.println("Enter total classes conducted: ");
8         int totalClasses = sc.nextInt();
9
10        System.out.println("Enter classes attended: ");
11        int attendedClasses = sc.nextInt();
12
13        if (totalClasses == 0) {
14            System.out.println("Invalid Data: Total classes cannot be zero.");
15        }
16        else if (attendedClasses > totalClasses || attendedClasses < 0) {
17            System.out.println("Invalid Data: Attended classes cannot be greater than total classes or negative.");
18        }
19        else {
20            double attendancePercentage = ((double) attendedClasses / totalClasses) * 100;
21            System.out.printf("Attendance Percentage: %.2f%%\n", attendancePercentage);
22
23            if (attendancePercentage >= 75) {
```

The Run console shows the following output:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=51192" -Dfile.encoding=UTF-8
Enter total classes conducted: 665
Enter classes attended: 555
Attendance Percentage: 83.46%
Eligible for Exam
```

The bottom status bar indicates the file encoding is UTF-8 and the project is 'Tutorial-2'.

AI Age Predictor



The screenshot shows the IntelliJ IDEA IDE with the 'AI Age Predictor' project open. The 'Main.java' file is displayed, containing the following code:

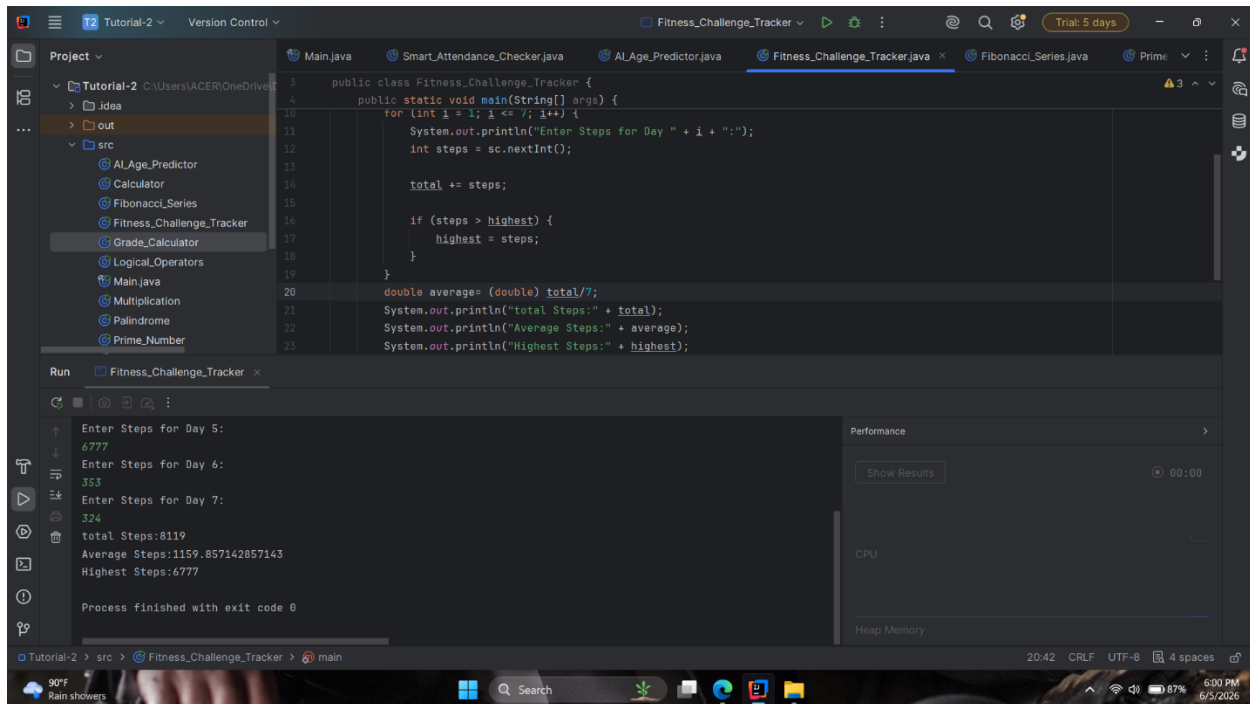
```
4 public class AI_Age_Predictor {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         System.out.println("Enter your current age: ");
9         int age = sc.nextInt();
10
11        System.out.println("Age after 10 years: " + (age + 10));
12        System.out.println("Age after 25 years: " + (age + 25));
13        System.out.println("Age after 50 years: " + (age + 50));
14
15        int currentYear = Year.now().getValue();
16        int yearTurn100 = currentYear + (100 - age);
17        {
18            System.out.println("You will turn 100 in the year: " + yearTurn100);
19        }
20        sc.close();
21    }
22 }
```

The Run console shows the following output:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=50404" -Dfile.encoding=UTF-8
Enter your current age: 20
Age after 10 years: 30
Age after 25 years: 45
Age after 50 years: 70
You will turn 100 in the year: 2106
Process finished with exit code 0
```

The bottom status bar indicates the file encoding is UTF-8 and the project is 'Tutorial-2'.

Fitness Challenge Tracker



```
public class Fitness_Challenge_Tracker {
    public static void main(String[] args) {
        for (int i = 1; i <= 7; i++) {
            System.out.println("Enter Steps for Day " + i + ":");
            int steps = sc.nextInt();

            total += steps;

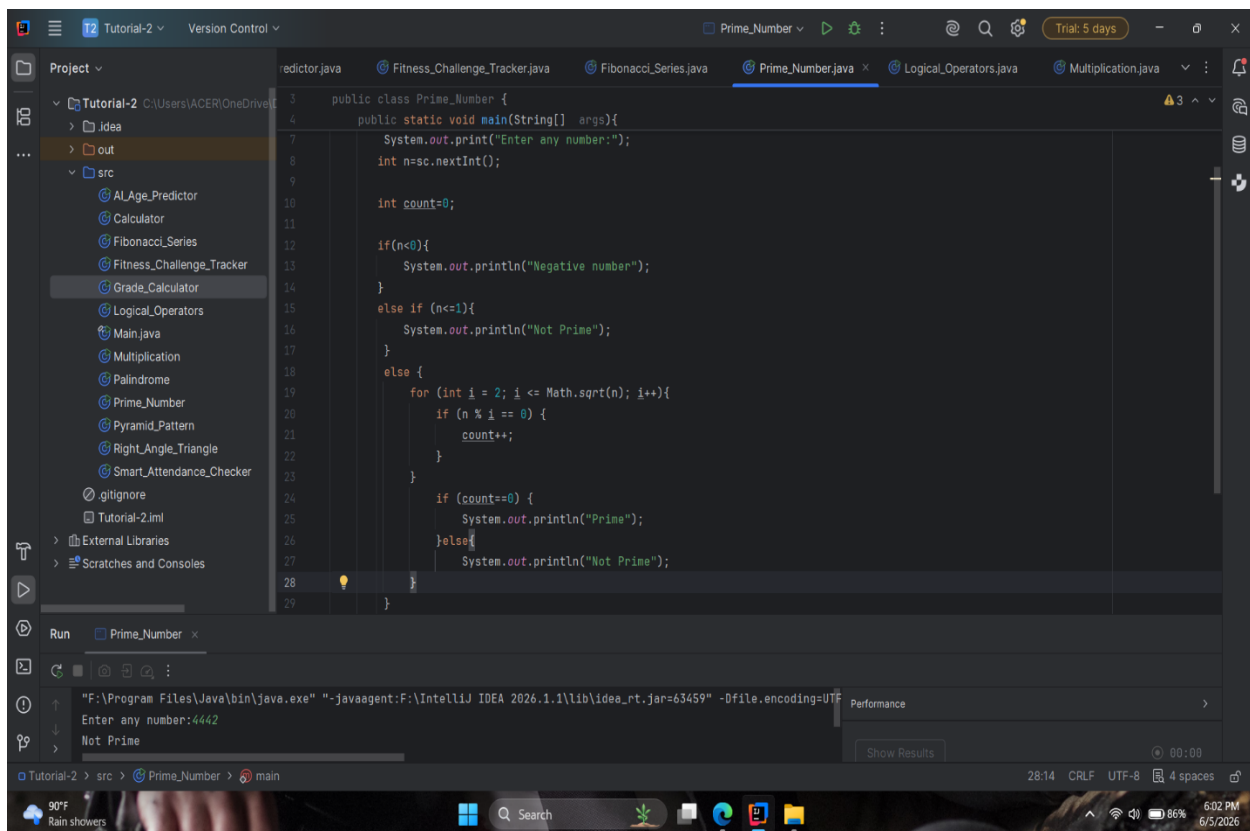
            if (steps > highest) {
                highest = steps;
            }
        }

        double average = (double) total / 7;
        System.out.println("Total Steps: " + total);
        System.out.println("Average Steps: " + average);
        System.out.println("Highest Steps: " + highest);
    }
}
```

Run Fitness_Challenge_Tracker

Enter Steps for Day 5:
6777
Enter Steps for Day 6:
353
Enter Steps for Day 7:
324
total Steps: 8119
Average Steps: 1159.857142857143
Highest Steps: 6777
Process finished with exit code 0

Prime Number



```
public class Prime_Number {
    public static void main(String[] args) {
        System.out.print("Enter any number:");
        int n = sc.nextInt();

        int count = 0;

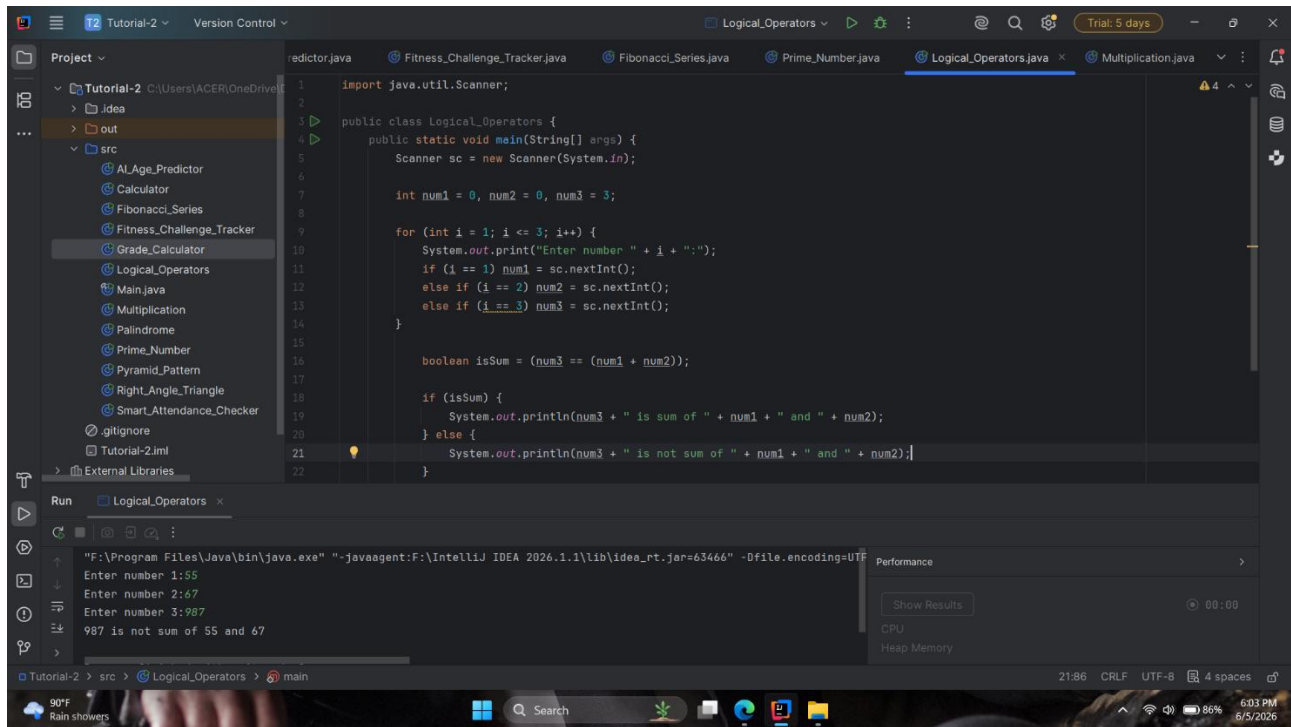
        if (n < 0) {
            System.out.println("Negative number");
        }
        else if (n < 1) {
            System.out.println("Not Prime");
        }
        else {
            for (int i = 2; i <= Math.sqrt(n); i++) {
                if (n % i == 0) {
                    count++;
                }
            }

            if (count == 0) {
                System.out.println("Prime");
            }
            else {
                System.out.println("Not Prime");
            }
        }
    }
}
```

Run Prime_Number

"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=63459" -Dfile.encoding=UTF-8
Enter any number: 4442
Not Prime

Logical Operators

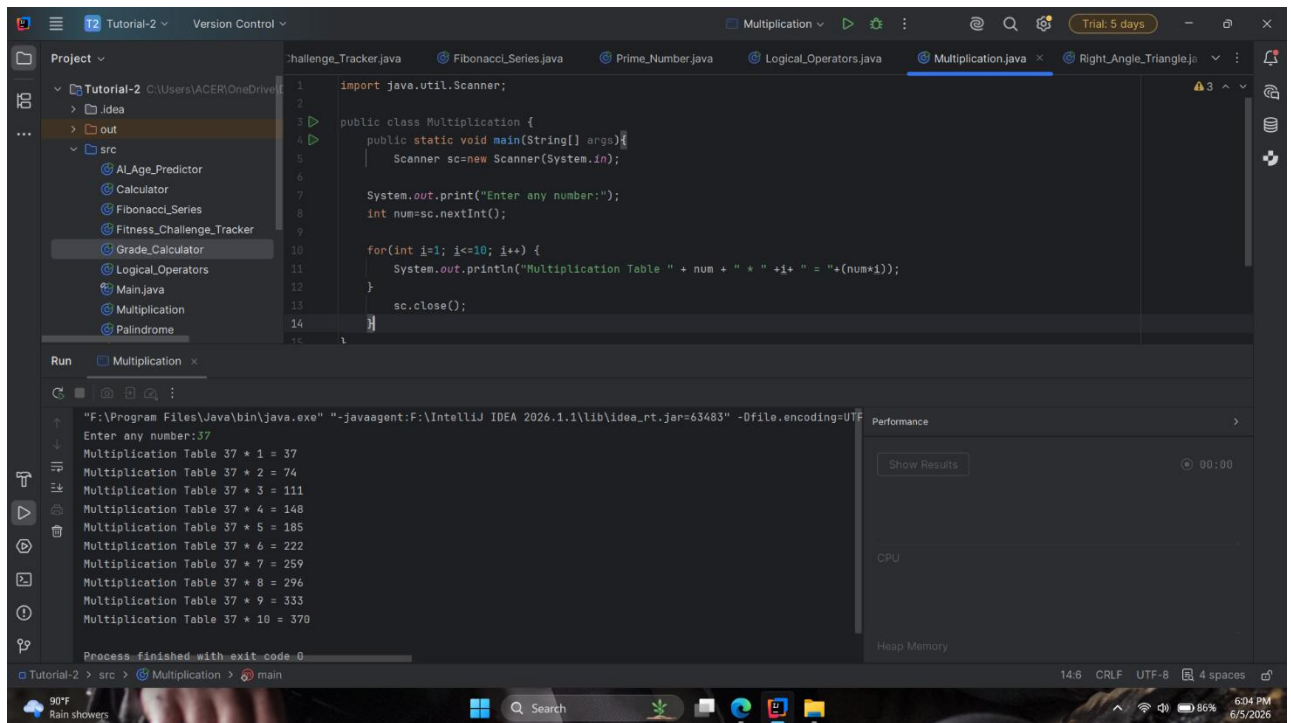


```
1 import java.util.Scanner;
2
3 public class Logical_Operators {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int num1 = 0, num2 = 0, num3 = 3;
8
9         for (int i = 1; i <= 3; i++) {
10             System.out.print("Enter number " + i + ":");
11             if (i == 1) num1 = sc.nextInt();
12             else if (i == 2) num2 = sc.nextInt();
13             else if (i == 3) num3 = sc.nextInt();
14         }
15
16         boolean isSum = (num3 == (num1 + num2));
17
18         if (isSum) {
19             System.out.println(num3 + " is sum of " + num1 + " and " + num2);
20         } else {
21             System.out.println(num3 + " is not sum of " + num1 + " and " + num2);
22         }
23     }
24 }
```

Run Console Output:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=63466" -Dfile.encoding=UTF-8
Enter number 1:55
Enter number 2:67
Enter number 3:987
987 is not sum of 55 and 67
```

Multiplication



```
1 import java.util.Scanner;
2
3 public class Multiplication {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter any number:");
8         int num = sc.nextInt();
9
10        for (int i = 1; i <= 10; i++) {
11            System.out.println("Multiplication Table " + num + " * " + i + " = " + (num * i));
12        }
13        sc.close();
14    }
15 }
```

Run Console Output:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=63483" -Dfile.encoding=UTF-8
Enter any number:37
Multiplication Table 37 * 1 = 37
Multiplication Table 37 * 2 = 74
Multiplication Table 37 * 3 = 111
Multiplication Table 37 * 4 = 148
Multiplication Table 37 * 5 = 185
Multiplication Table 37 * 6 = 222
Multiplication Table 37 * 7 = 259
Multiplication Table 37 * 8 = 296
Multiplication Table 37 * 9 = 333
Multiplication Table 37 * 10 = 370
Process finished with exit code 0
```

Right Angle triangle

The screenshot shows the IntelliJ IDEA IDE with the 'Right_Angle_Triangle.java' file open. The code defines a class with a main method that uses a Scanner to take input from the user. It then uses nested loops to print a right-angled triangle of asterisks. The output window shows the program running successfully, displaying the triangle for 6 rows.

```
1 import java.util.Scanner;
2
3 public class Right_Angle_Triangle {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.println("Enter number of rows: ");
8         int rows = sc.nextInt();
9
10        for(int i=1; i<=rows; i++) {
11            for (int j = 1; j <= i; j++) {
12                System.out.print("*");
13            }
14            System.out.println("");
15        }
16        sc.close();
17    }
18 }
```

Run Output:

```
Enter number of rows:
6
*
**
***
****
*****
*****
Process finished with exit code 0
```

Pyramid Pattern

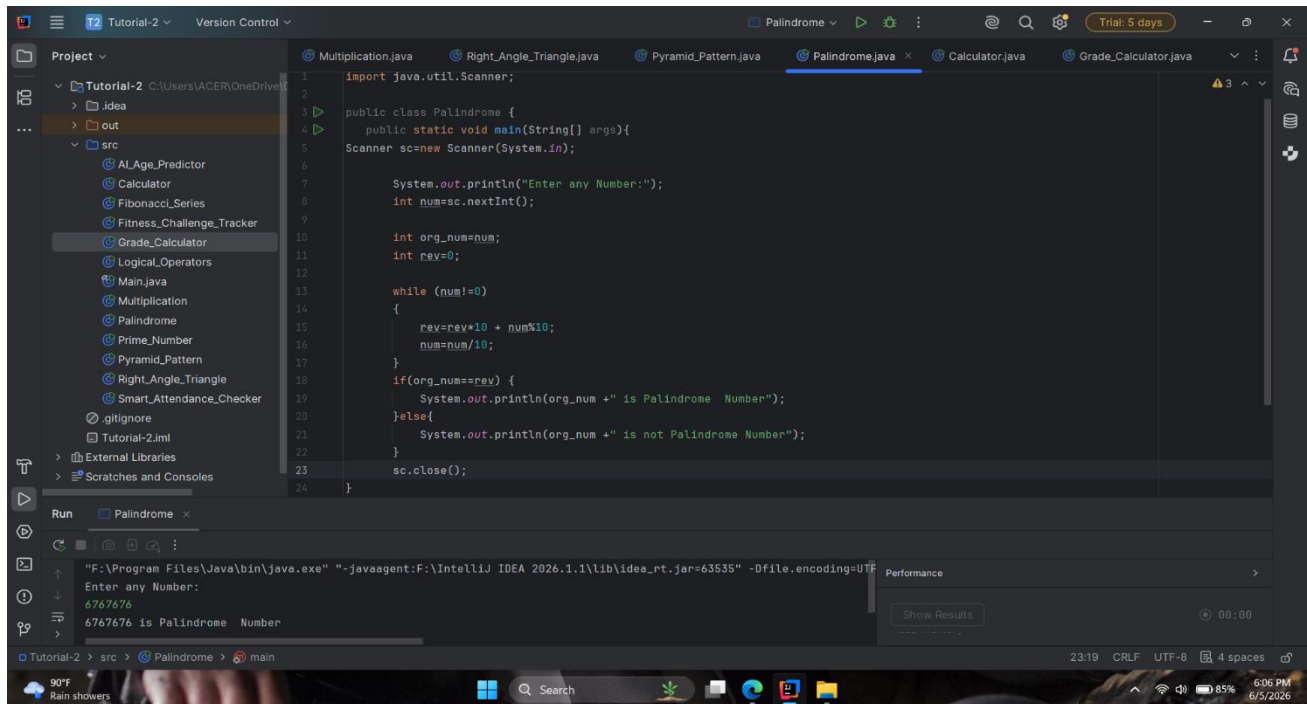
The screenshot shows the IntelliJ IDEA IDE with the 'Pyramid_Pattern.java' file open. The code defines a class with a main method that uses a Scanner to take input from the user. It then uses nested loops to print a pyramid pattern of asterisks. The output window shows the program running successfully, displaying the pyramid for 7 rows.

```
1 import java.util.Scanner;
2
3 public class Pyramid_Pattern {
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         System.out.println("Enter any number of rows: ");
8         int rows=sc.nextInt();
9
10        for(int i=1; i<=rows; i++) {
11            for (int j = 1; j <= rows; j++) {
12                System.out.print(" ");
13            }
14            for (int k = 1; k<=(2*i-1); k++) {
15                System.out.print("*");
16            }
17            System.out.println();
18        }
19        sc.close();
20    }
21 }
```

Run Output:

```
Enter any number of rows:
7
      *
     ***
    *****
   *****
  *****
 *****
*****
Process finished with exit code 0
```

Palindrome Number



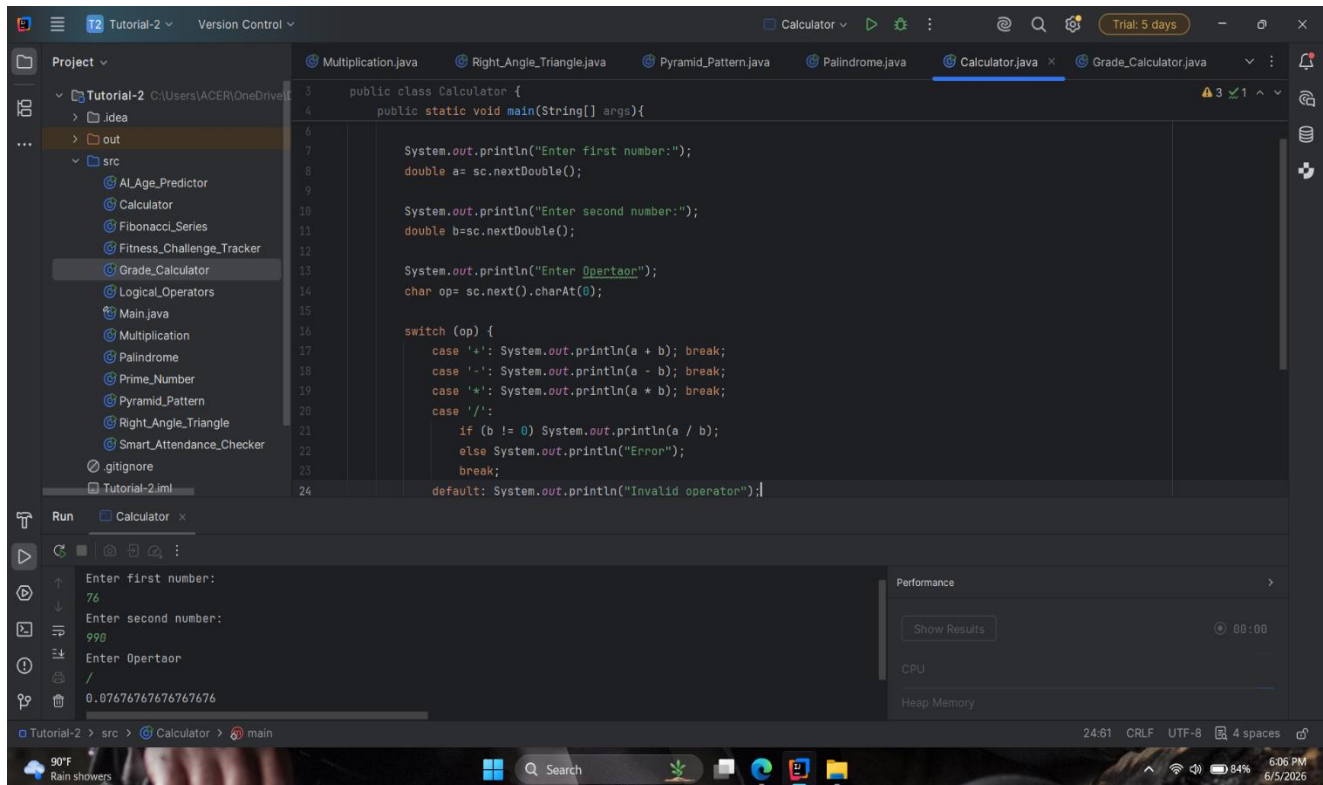
The screenshot shows an IDE with the following components:

- Project Explorer:** Lists various Java projects including `ALAge_Predictor`, `Calculator`, `Fibonacci_Series`, `Fitness_Challenge_Tracker`, `Grade_Calculator`, `Logical_Operators`, `Main.java`, `Multiplication`, `Palindrome`, `Prime_Number`, `Pyramid_Pattern`, `Right_Angle_Triangle`, and `Smart_Attendance_Checker`.
- Code Editor:** Displays the `Palindrome.java` file with the following code:

```
1 import java.util.Scanner;
2
3 public class Palindrome {
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         System.out.println("Enter any Number:");
8         int num=sc.nextInt();
9
10        int org_num=num;
11        int rev=0;
12
13        while (num!=0)
14        {
15            rev=rev*10 + num%10;
16            num=num/10;
17        }
18        if(org_num==rev) {
19            System.out.println(org_num + " is Palindrome Number");
20        }else{
21            System.out.println(org_num + " is not Palindrome Number");
22        }
23        sc.close();
24    }
```
- Run Console:** Shows the execution output:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=63535" -Dfile.encoding=UTF-8
Enter any Number:
6767676
6767676 is Palindrome Number
```
- Performance:** A section on the right showing CPU and Heap Memory usage.

Calculator



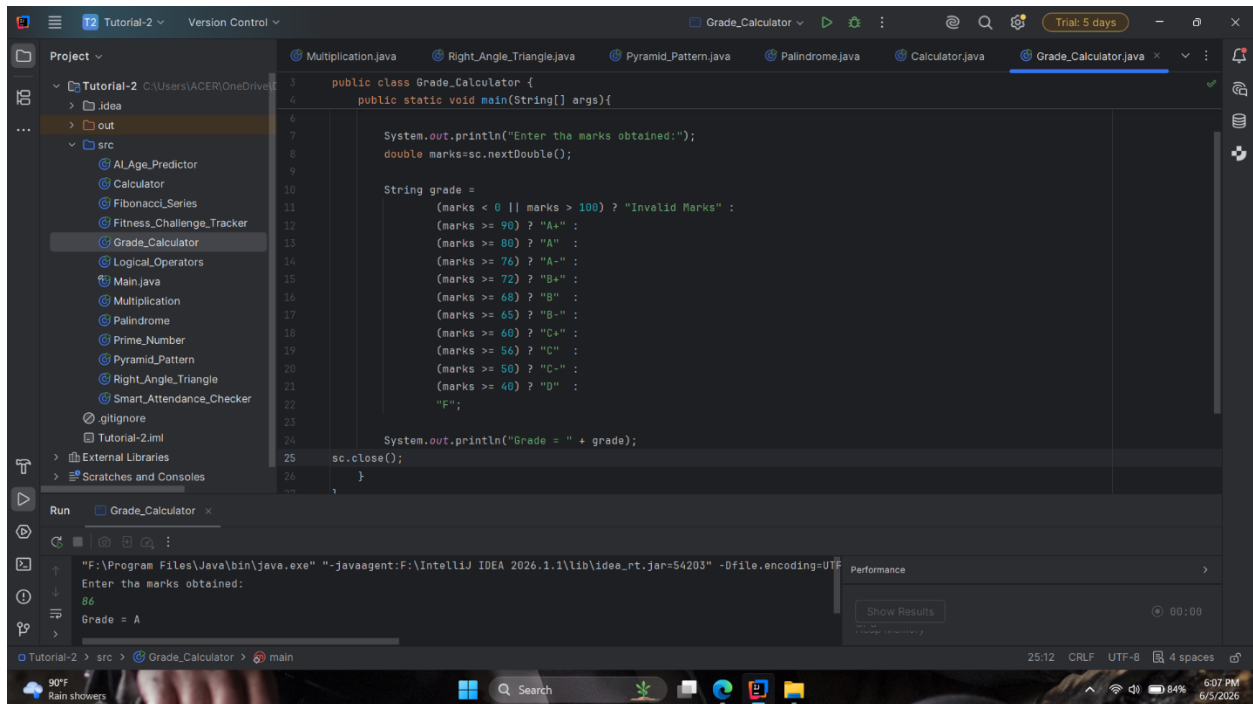
The screenshot shows an IDE with the following components:

- Project Explorer:** Lists various Java projects including `ALAge_Predictor`, `Calculator`, `Fibonacci_Series`, `Fitness_Challenge_Tracker`, `Grade_Calculator`, `Logical_Operators`, `Main.java`, `Multiplication`, `Palindrome`, `Prime_Number`, `Pyramid_Pattern`, `Right_Angle_Triangle`, and `Smart_Attendance_Checker`.
- Code Editor:** Displays the `Calculator.java` file with the following code:

```
3 public class Calculator {
4     public static void main(String[] args){
5
6         System.out.println("Enter first number:");
7         double a= sc.nextDouble();
8
9         System.out.println("Enter second number:");
10        double b=sc.nextDouble();
11
12        System.out.println("Enter Operator");
13        char op= sc.next().charAt(0);
14
15        switch (op) {
16            case '+': System.out.println(a + b); break;
17            case '-': System.out.println(a - b); break;
18            case '*': System.out.println(a * b); break;
19            case '/':
20                if (b != 0) System.out.println(a / b);
21                else System.out.println("Error");
22                break;
23            default: System.out.println("Invalid operator");
24        }
```
- Run Console:** Shows the execution output:

```
Enter first number:
76
Enter second number:
999
Enter Operator
/
0.07676767676767676
```
- Performance:** A section on the right showing CPU and Heap Memory usage.

Grade Calculator



Fibonacci Series

